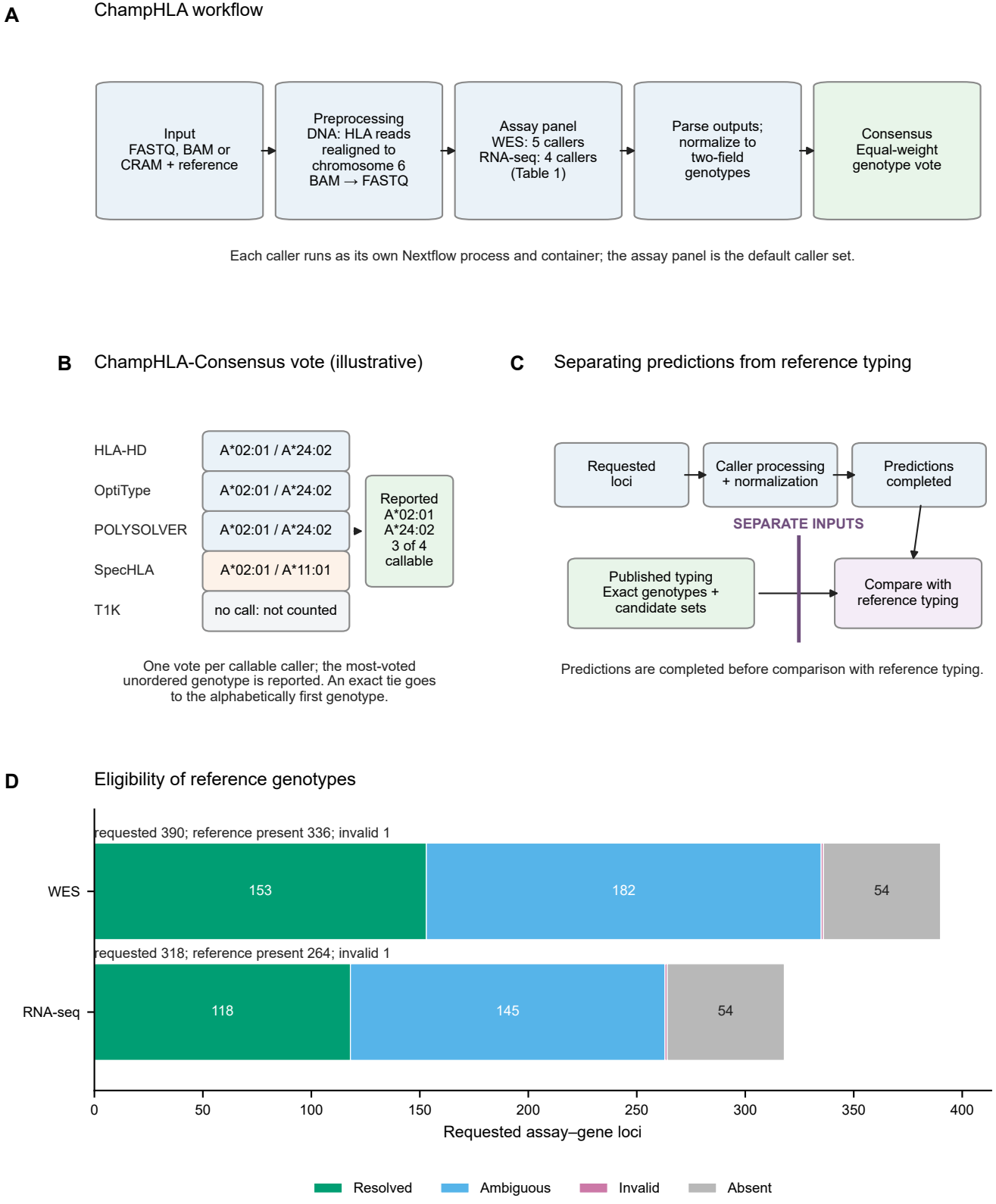

A
ChampHLA workflow
Input
FASTQ, BAM or
CRAM + reference
Preprocessing
DNA: HLA reads
realigned to
chromosome 6
BAM → FASTQ
Assay panel
WES: 5 callers
RNA-seq: 4 callers
(Table 1)
Parse outputs;
normalize to
two-field
genotypes
Consensus
Equal-weight
genotype vote
Each caller runs as its own Nextflow process and container; the assay panel is the default caller set.
B
ChampHLA-Consensus vote (illustrative)
HLA-HD
A*02:01 / A*24:02
OptiType
A*02:01 / A*24:02
POLYSOLVER
A*02:01 / A*24:02
SpecHLA
A*02:01 / A*11:01
T1K
no call: not counted
Reported
A*02:01
A*24:02
3 of 4
callable
One vote per callable caller; the most-voted
unordered genotype is reported. An exact tie goes
to the alphabetically first genotype.
C
Separating predictions from reference typing
Requested
loci
Caller processing
+ normalization
Predictions
completed
SEPARATE INPUTS
Published typing
Exact genotypes +
candidate sets
Compare with
reference typing
Predictions are completed before comparison with reference typing.
D
Eligibility of reference genotypes
requested 390; reference present 336; invalid 1
WES
153
182
54
requested 318; reference present 264; invalid 1
RNA-seq
118
145
54
0
50
100
150
200
250
300
350
400
Requested assay–gene loci
Resolved
Ambiguous
Invalid
Absent